

**Cambridge (CIE) IGCSE Maths
Paper 2: Non-Calculator (Set A)
Extended**

Saturday 26 April 2025

Morning (Time: 2 hours 0 minutes)

Total marks

/ 101

Instructions

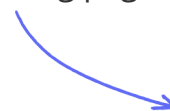
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

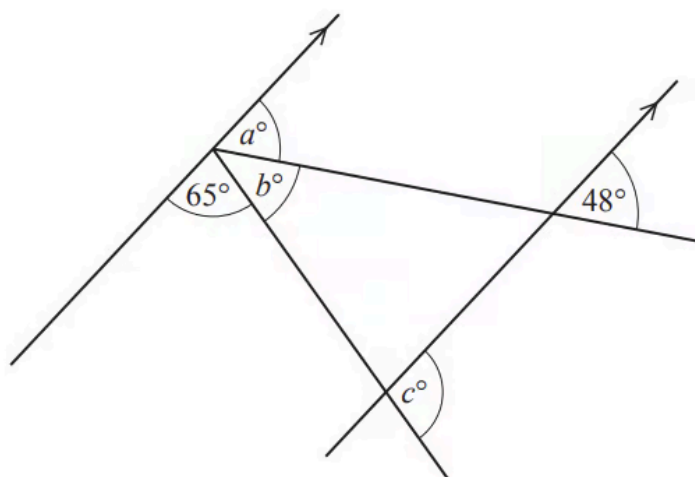
- [List of formulas](#)

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or visit the mock exams landing page for this course



1 (a)



NOT TO
SCALE

The diagram shows two parallel lines and two straight lines.

Find the value of a .

Give a reason for your answer.

$a = \dots\dots\dots$ because $\dots\dots\dots$ [2]

(2 marks)

(b) Find the value of b .

Give a reason for your answer.

$b = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

(c) Find the value of c .

$c = \dots\dots\dots$

(2 marks)

2 (a) Factorise $12x + 15$

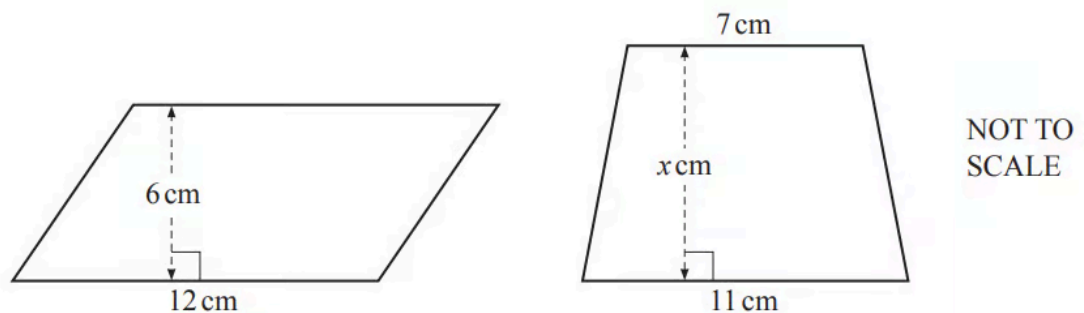
(1 mark)

(b) $7x - 3 = 2x + 11$

$x = \dots\dots\dots$

(2 marks)

3



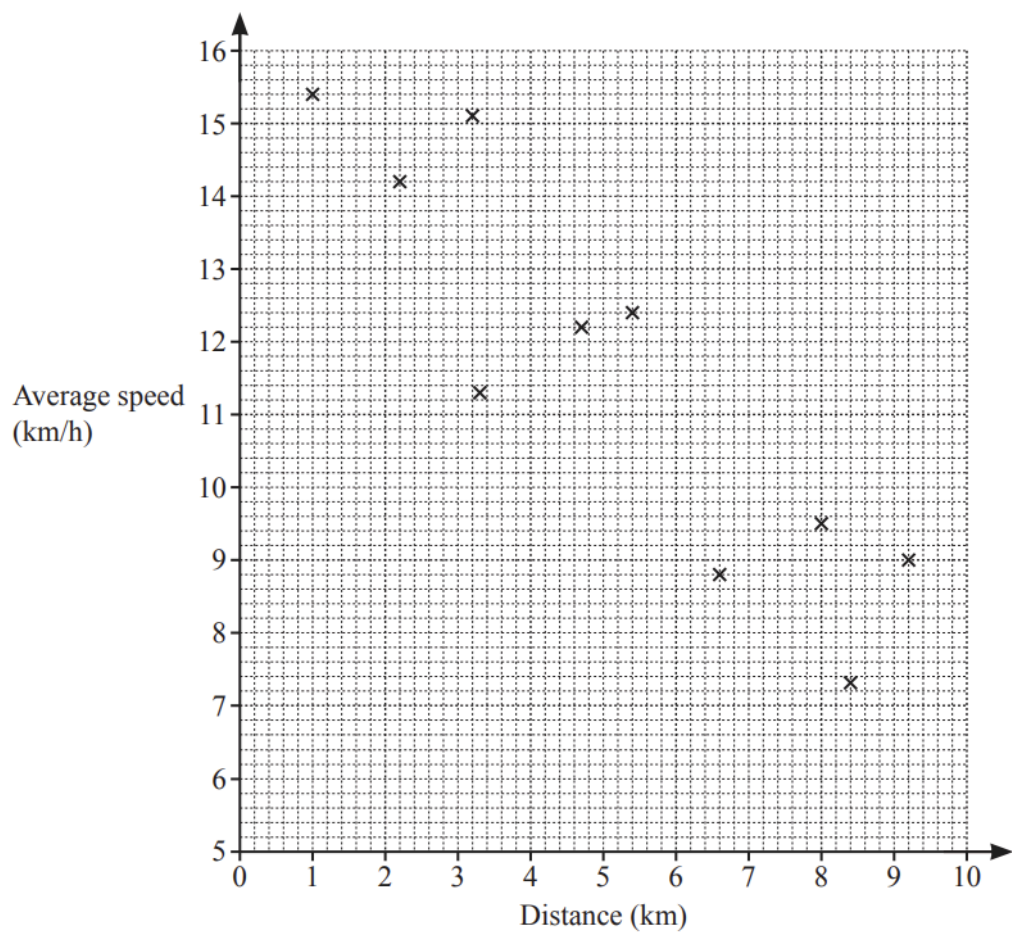
The area of the parallelogram is the same as the area of the trapezium.

Work out the value of x .

$x = \dots\dots\dots$

(3 marks)

4 (a) Aisha records the distance she runs and her average speed. The results are shown in the scatter diagram.



The table shows the results of four more runs.

Distance (km)	4.2	5.7	7.1	8.8
Average speed (km/h)	13.4	11.8	9.8	8.3

On the scatter diagram, plot these points.

(2 marks)

(b) What type of correlation is shown in the scatter diagram?

(1 mark)

(c) On the scatter diagram, draw a line of best fit.

(1 mark)

(d) Use your line of best fit to estimate her average speed when she runs a distance of 6km.

.....km/h

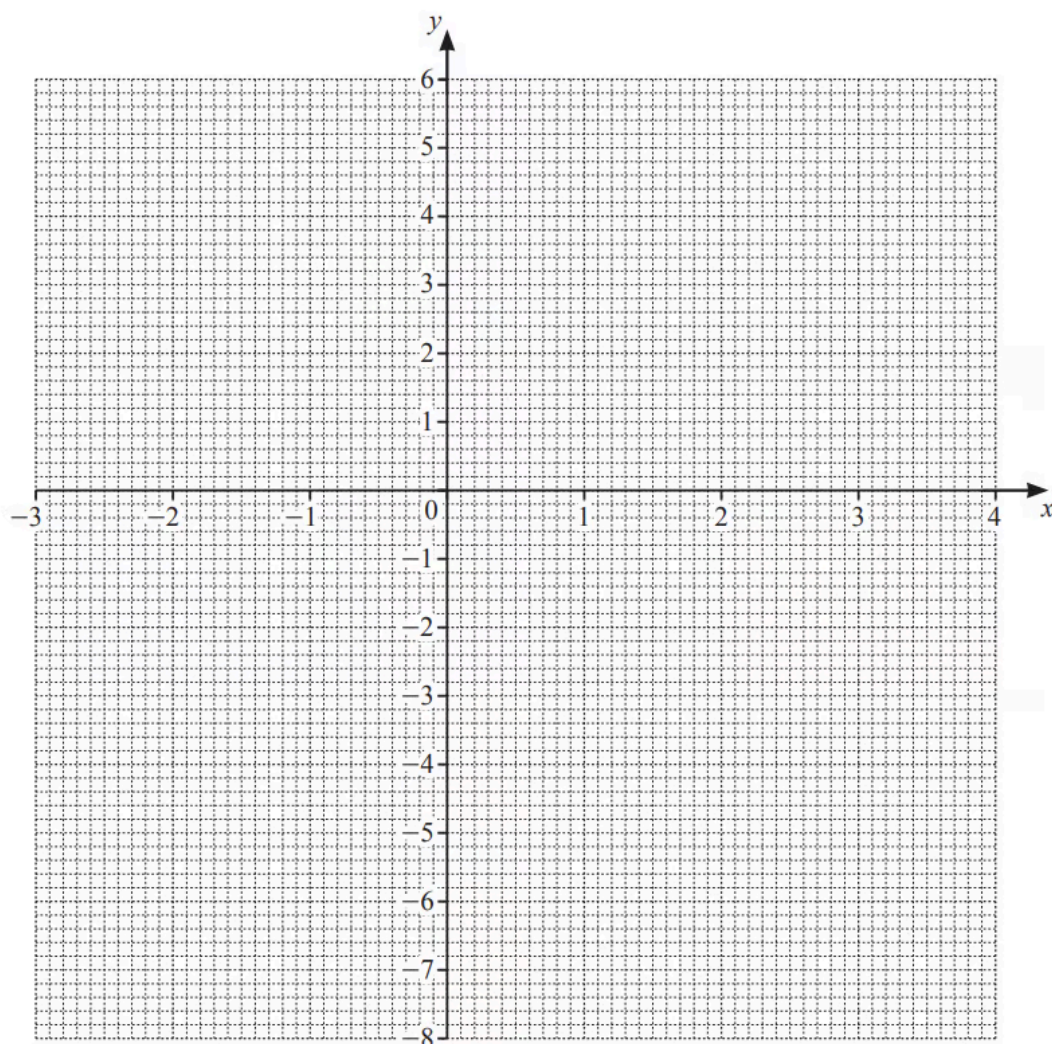
(1 mark)

5 (a) Complete the table of values for $y = -x^2 + x + 5$

x	-3	-2	-1	0	1	2	3	4
y		-1	3			3		

(3 marks)

(b) On the grid, draw the graph of $y = -x^2 + x + 5$ for $-3 \leq x \leq 4$.



(4 marks)

(c) Write down the coordinates of the highest point of the graph.

(1 mark)

(d) Write down the equation of the line of symmetry of the graph.

(1 mark)

(e) i) On the grid, draw the line $y = x$ for $-3 \leq x \leq 4$

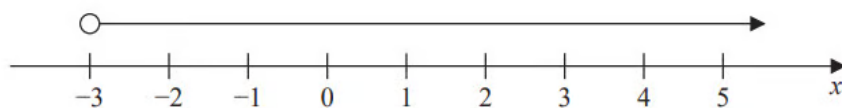
[1]

ii) Write down the values of x where the line $y = x$ crosses the curve $y = -x^2 + x + 5$

$x = \dots\dots\dots$ and $x = \dots\dots\dots$ [2]

(3 marks)

6 (a)



Write down the inequality shown on the number line.

(1 mark)

(b) List all of the integers which satisfy **all three** of the following inequalities.

$$x \leq 8$$

$$x > 2$$

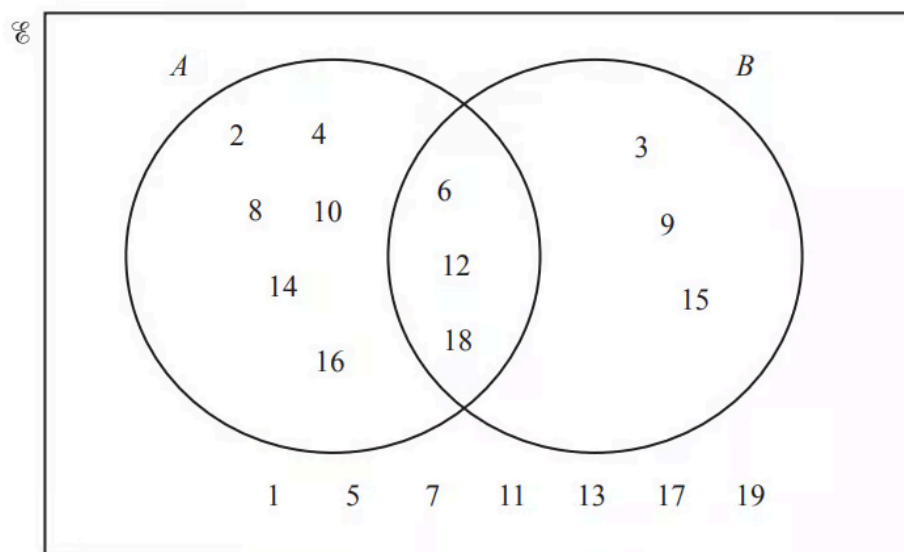
$$x > 3$$

(1 mark)

7 (a) $\mathcal{E} = \{x : x \text{ is a positive integer less than } 20\}$

$A = \{x : x \text{ is an even number}\}$

$B = \{x : x \text{ is a multiple of } 3\}$



Write down $n(A)$.

(1 mark)

(b) List the elements of set B .

$B = \{ \dots \}$

(2 marks)

(c) Complete the statement.

$A \cap B = \{x : x \text{ is } \dots\}$

(1 mark)

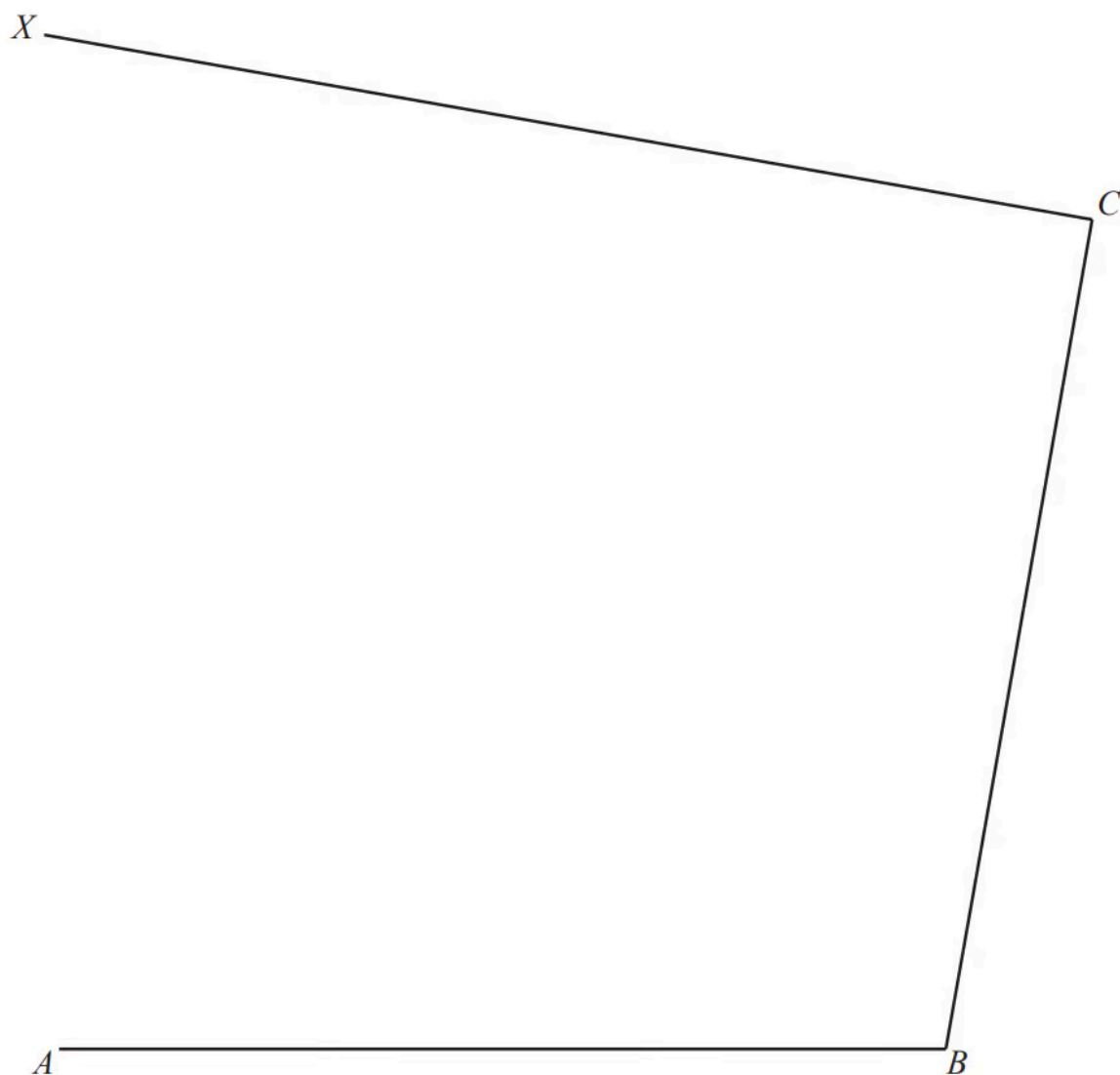
- 8 The height, h metres, of a tower is 76.3 m, correct to 1 decimal place.
Complete this statement about the value of h .

..... $\leq h <$

(2 marks)

- 9 (a)** The diagram shows an incomplete scale drawing of a market place, $ABCD$, where D is on CX .

The scale is 1 centimetre represents 5 metres.



Scale : 1 cm to 5 m

D lies on CX such that angle $DAB = 75^\circ$.

On the diagram, draw the line AD and mark the position of D .

(2 marks)

(b) Find the actual length of the side BC of the market place.

..... m

(2 marks)

(c) Write the scale of the drawing in the form $1 : n$.

1 :

(1 mark)

10 $m = \sqrt{\frac{k+1}{4}}$

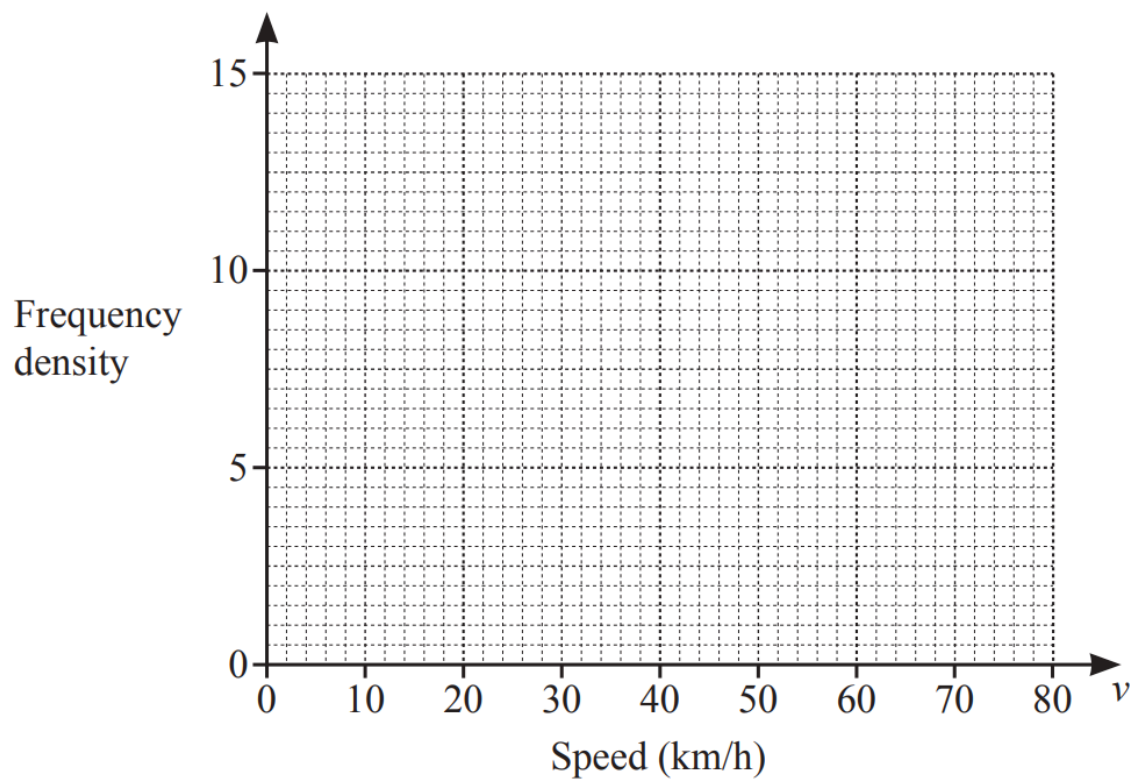
11. Make k the subject of the formula.

(3 marks)

11 The speed, v km/h, of each of 200 cars passing a building is measured. The frequency table shows the results.

Speed (v km/h)	$0 < v \leq 40$	$40 < v \leq 50$	$50 < v \leq 80$
Frequency	50	120	30

On the grid, draw a histogram to show the information in this table.

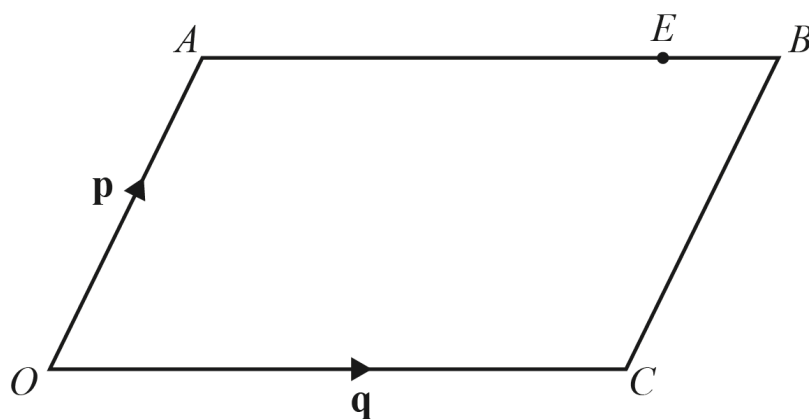


(3 marks)

12 Rationalise the denominator and simplify $\frac{10}{3\sqrt{5}}$

(2 marks)

13

NOT TO
SCALE

a parallelogram.

$$\vec{OA} = \mathbf{p} \text{ and } \vec{OC} = \mathbf{q}.$$

E is the point on AB such that $AE : EB = 3 : 1$.

Find $\vec{OE}$, in terms of $\mathbf{p}$ and $\mathbf{q}$, in its simplest form.

$$\vec{OE} = \dots\dots\dots$$

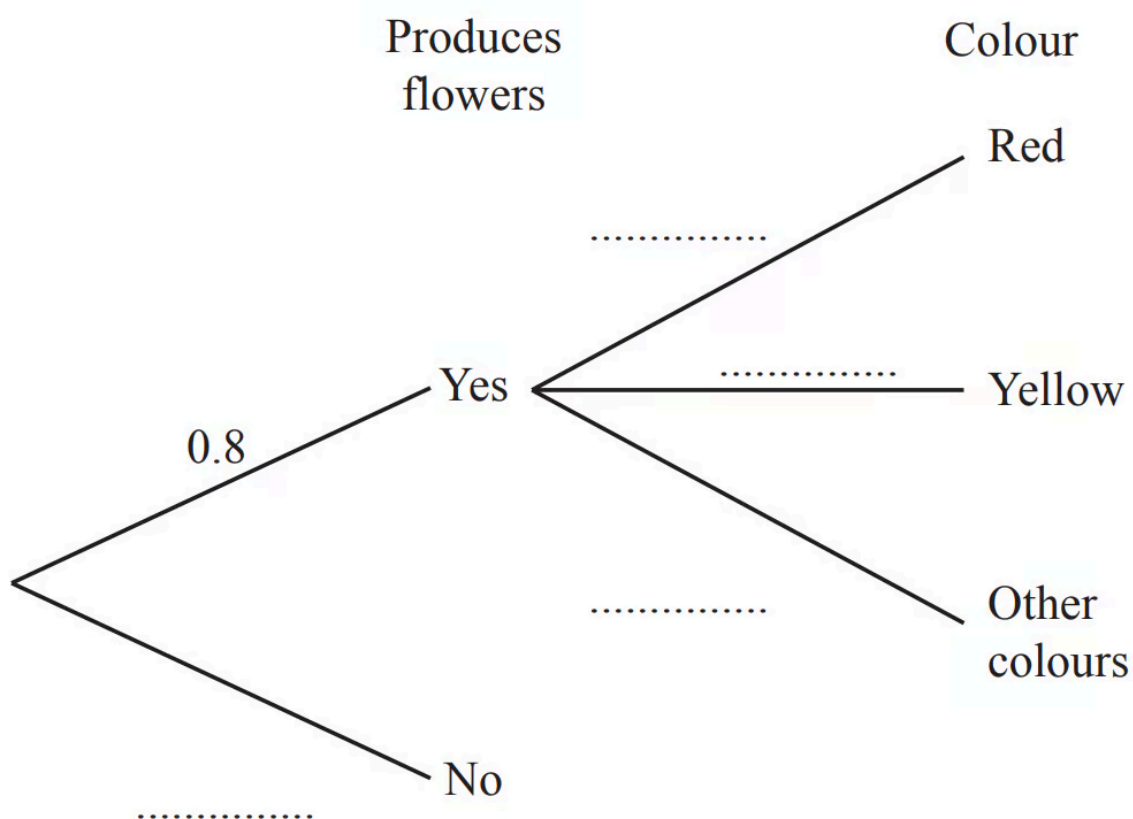
(2 marks)

14 (a) Tanya plants some seeds.

The probability that a seed will produce flowers is 0.8.

When a seed produces flowers, the probability that the flowers are red is 0.6 and the probability that the flowers are yellow is 0.3.

Complete the tree diagram.



(2 marks)

(b) Find the probability that a seed chosen at random produces red flowers.

(2 marks)

- (c) Tanya chooses a seed at random.

Find the probability that this seed does not produce red flowers and does not produce yellow flowers.

(3 marks)

- (d) Two of the seeds are chosen at random.

Find the probability that one produces flowers and one does not produce flowers.

(3 marks)

- 15 A is the point $(3, 5)$ and B is the point $(1, -7)$.

Find the equation of the line perpendicular to AB that passes through the point A .

Give your answer in the form $y = mx + c$.

$y =$

(4 marks)

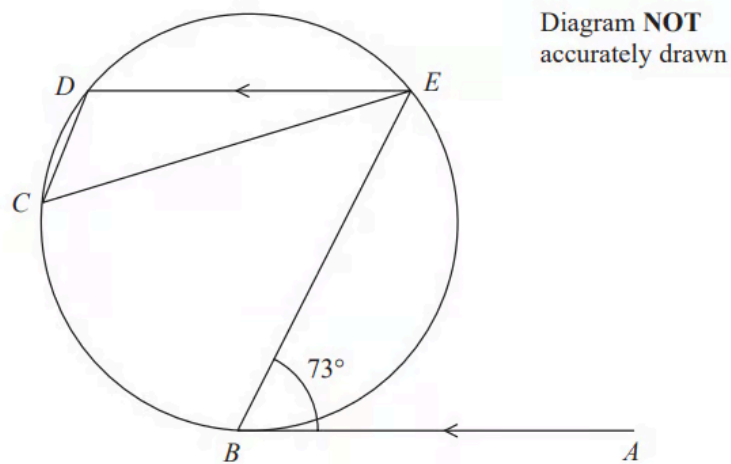
- 16 $y = x^4 - 4x^3$

Find the two stationary points on the graph of $y = x^4 - 4x^3$.

(..... ,)
(..... ,)

(6 marks)

17



B , C , D and E are points on a circle.

AB is the tangent at B to the circle.

AB is parallel to ED .

Angle $ABE = 73^\circ$

Work out the size of angle DCE .

Give a reason for each stage of your working.

(5 marks)

18

$$f(x) = 8 - 3x \quad g(x) = \frac{10}{x+1}, x \neq -1 \quad h(x) = 2^x$$

Find

i) $hf\left(\frac{8}{3}\right),$

[2]

ii) $gh(-2),$

[2]

iii) $g^{-1}(x),$

$$g^{-1}(x) = \dots\dots\dots [3]$$

iv) $f^{-1}f(5).$

[1]

(8 marks)

19

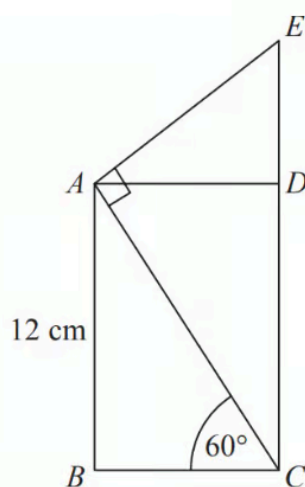


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CDE is a straight line.

$AB = 12$ cm

Angle $ACB = 60^\circ$

Angle $EAC = 90^\circ$

Calculate the length of CE .

You must show all your working.

(4 marks)

20 Solve algebraically the simultaneous equations

$$x^2 + y^2 = 25$$

$$y - 2x = 5$$

(5 marks)

21 ABCD is a trapezium.



Not to scale

The perimeter of the trapezium is 56 cm. The ratio $AD : AB : DC : BC = 5 : 12 : 6 : 5$.

Calculate the area of the trapezium.

Show your working.

.....cm²

(7 marks)